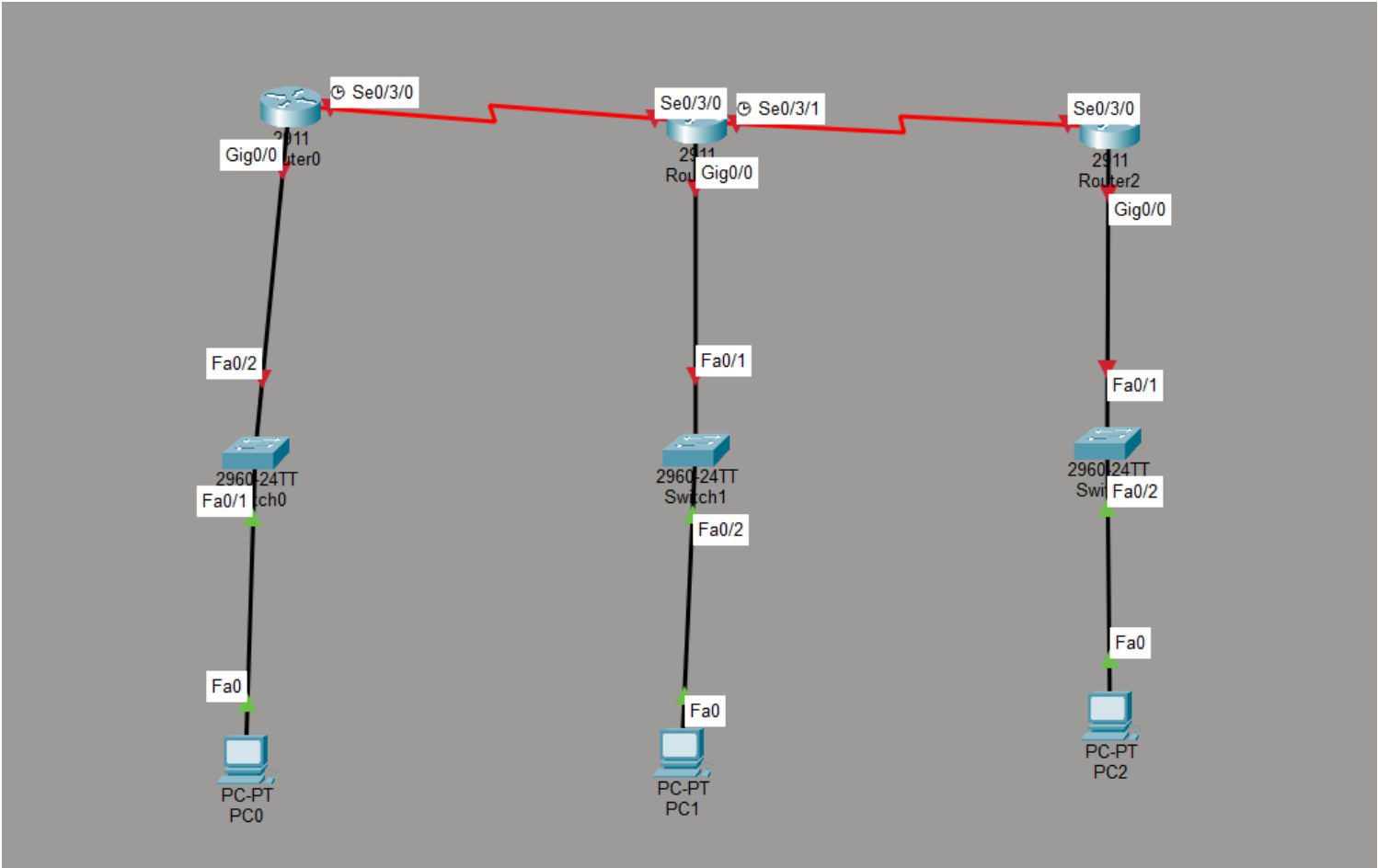


**Q-Enable RIP version 2 on all three routers in your topology, advertise all directly connected networks, and verify that each router learns routes to all subnets using the 'show ip route' command.**

To configure a three-router network in Cisco Packet Tracer, assign IPv4 addresses to router and PC interfaces, connect routers using serial links, enable RIP version 2 on all routers, advertise all directly connected networks, and verify learned routes using show ip route.

**Network Topology**



**IP Addressing Table**

| Device  | Interface | IP Address   | Subnet Mask     | Purpose             |
|---------|-----------|--------------|-----------------|---------------------|
| Router0 | G0/0      | 192.168.1.1  | 255.255.255.0   | LAN 1 gateway       |
| Router0 | S0/3/0    | 10.0.0.1     | 255.255.255.252 | Router0-Router1     |
| Router1 | G0/0      | 192.168.2.1  | 255.255.255.0   | LAN 2 gateway       |
| Router1 | S0/3/0    | 10.0.0.2     | 255.255.255.252 | Router1-Router0     |
| Router1 | S0/3/1    | 10.0.0.5     | 255.255.255.252 | Router1-Router2     |
| Router2 | G0/0      | 192.168.3.1  | 255.255.255.0   | LAN 3 gateway       |
| Router2 | S0/3/0    | 10.0.0.6     | 255.255.255.252 | Router2-Router1     |
| PC0     | Fa0       | 192.168.1.10 | 255.255.255.0   | Gateway 192.168.1.1 |
| PC1     | Fa0       | 192.168.2.10 | 255.255.255.0   | Gateway 192.168.2.1 |
| PC2     | Fa0       | 192.168.3.10 | 255.255.255.0   | Gateway 192.168.3.1 |

# RIP Version 2 Configuration

## Router 0

```
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#no auto-summary
Router(config-router)#network 192.168.1.0
Router(config-router)#network 10.0.0.0
Router(config-router)#exit
Router(config)#do wr
Building configuration...
[OK]
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#
```

## Router 1

```
Router(config)#
Router(config)#router rip
Router(config-router)#
Router(config-router)#version 2
Router(config-router)#no auto-summary
Router(config-router)#network 192.168.2.0
Router(config-router)#network 10.0.0.0
Router(config-router)#network 10.0.0.4
Router(config-router)#exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

## Router 2

```
Router(config)#
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#no auto-summary
Router(config-router)#network 192.168.3.0
Router(config-router)#network 10.0.0.4
Router(config-router)#exit
Router(config)#do wr
Building configuration...
[OK]
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

## Verification

Use these commands after RIP has converged:

- show ip interface brief
- show ip route
- show ip protocols

In show ip route, C indicates a directly connected network and R indicates a route learned through RIP.

| Router  | RIP-learned networks expected               | Verification |
|---------|---------------------------------------------|--------------|
| Router0 | 192.168.2.0/24, 10.0.0.4/30, 192.168.3.0/24 | R entries    |

|         |                                                   |           |
|---------|---------------------------------------------------|-----------|
| Router1 | 192.168.1.0/24,<br>192.168.3.0/24                 | R entries |
| Router2 | 10.0.0.0/30,<br>192.168.1.0/24,<br>192.168.2.0/24 | R entries |

Connectivity Testing

PC 0

```
C:\>ping 192.168.3.10

Pinging 192.168.3.10 with 32 bytes of data:

Reply from 192.168.3.10: bytes=32 time=9ms TTL=125
Reply from 192.168.3.10: bytes=32 time=2ms TTL=125
Reply from 192.168.3.10: bytes=32 time=3ms TTL=125
Reply from 192.168.3.10: bytes=32 time=2ms TTL=125

Ping statistics for 192.168.3.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 9ms, Average = 4ms

C:\>
```

RIP version 2 was enabled on all three routers and all directly connected LAN and serial networks were advertised. After convergence, remote networks are learned dynamically and appear with R in the routing table.